

Question 2 – Employee Payroll Interface

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Objective / Aim

To implement an Employee Payroll System using a Java interface and demonstrate different salary details for regular and contract employees.

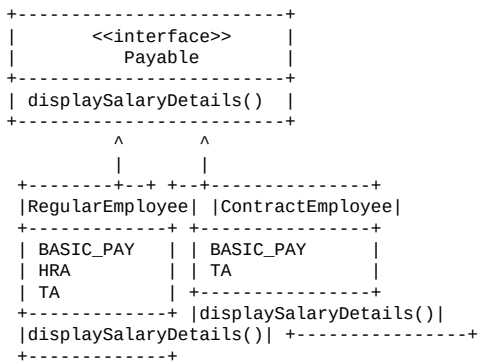
Algorithm / Logic

1. Create the Payable interface with displaySalaryDetails().
2. Create RegularEmployee implementing Payable.
3. Define basic pay, HRA, and TA for a regular employee.
4. Create ContractEmployee implementing Payable.
5. Define basic pay and TA for a contract employee with HRA as zero.
6. Read the employee type from the user.
7. Create the appropriate employee object using the Payable reference.
8. Display the salary details.

Steps for Execution

1. Create the Java project and add the Payable, RegularEmployee, ContractEmployee, and main class.
2. Compile and run Week8EmployeePayrollInterface.java.
3. Enter R for Regular or C for Contract.
4. The program creates the corresponding employee object.
5. Observe the salary components and total amount.

UML Class Diagram



Source Code

```
import java.util.Scanner;

interface Payable {
    void displaySalaryDetails();
}

class RegularEmployee implements Payable {
    private static final int BASIC_PAY = 25000;
    private static final int HRA = 15000;
    private static final int TA = 5000;
    public void displaySalaryDetails() {
        System.out.printf("Basic Pay: %d\nHRA: %d\nTA: %d\nTotal Amount: %d\n",
            BASIC_PAY, HRA, TA, BASIC_PAY + HRA + TA);
    }
}

class ContractEmployee implements Payable {
    private static final int BASIC_PAY = 12000;
    private static final int TA = 3000;
    public void displaySalaryDetails() {
        System.out.printf("Basic Pay: %d\nHRA: 0\nTA: %d\nTotal Amount: %d\n",
            BASIC_PAY, TA, BASIC_PAY + TA);
    }
}

public class Week8EmployeePayrollInterface {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter employee ID (R for Regular, C for Contract): ");
        String type = scanner.nextLine().trim().toUpperCase();
        Payable employee = type.startsWith("C") ? new ContractEmployee() : new RegularEmployee();
        employee.displaySalaryDetails();
        scanner.close();
    }
}
```

Output

Enter employee ID (R for Regular, C for Contract): R
Basic Pay: 25000
HRA: 15000
TA: 5000
Total Amount: 45000